GeoPath AI: Web-Based AI Math Differentiator

Selected topic: Pythagorean theorem and real-world right-triangle modelling. **Shared objective:** Apply $a^2 + b^2 = c^2$ accurately and justify the geometric interpretation.

Application overview. GeoPath AI is a responsive HTML/CSS/JavaScript website with three selectable learning pathways, a labelled geometric diagram, instant answer checking, level-specific hints, verified solutions, and varied numerical examples. Core exercises run locally without internet. An optional Gemini API integration uses a Netlify serverless function so the API key is never exposed in browser code.

How differentiation is achieved. Foundation uses simple numerical right triangles and explicit step-by-step scaffolding. Standard requires translating perpendicular park paths into a right-triangle model and explaining the diagonal. Advanced combines the diagonal of a rectangular garden with a material constraint, requiring calculation, comparison, and a justified design decision. All levels preserve the same mathematical concept while increasing interpretation and cognitive demand.

AI prompt design. The AI coach is instructed to act as a patient secondary mathematics teacher. Each request supplies the selected readiness level, the current problem, the teacher-verified solution, and the learner question. The prompt specifies: “For foundation give one scaffolded step at a time; for standard encourage modelling and justification; for advanced emphasize constraints and reasoning.” It also instructs the model to avoid revealing a complete solution unless requested, respect the verified reference, and acknowledge uncertainty.

Feedback, verification and limitations. Deterministic JavaScript checks numerical answers immediately and provides accurate worked examples for preselected Pythagorean triples. The AI coach can give personalized natural-language explanations only after a server-side API key is configured and the site is deployed. Otherwise the interface clearly identifies its local non-AI fallback. Teacher review of AI-generated responses remains necessary for mathematical and pedagogical accuracy.

Submission: Publish the site on Netlify, upload the source code and README to a public GitHub repository, and submit both links with this report. Replace this line with your links before submitting.